AZURE + FINOPS REFERENCE

Industry Best Practices: App Service Plans, Web Apps, and Container Registries

Optimize platform services through measured scaling, plan density, tier selection, image hygiene, and cost ownership. Preserve application availability, deployment safety, and supply-chain controls.

Applicable resource types

Microsoft.Web/serverfarms, Microsoft.Web/sites, and Microsoft.ContainerRegistry/registries.

Cost optimization practices

For App Service, evaluate the plan SKU, worker count, autoscale limits, and app-to-plan placement together. A plan is billed for provisioned capacity, so idle workers and fragmented plans require workload-level review.

• Use metric- and schedule-based autoscale with tested minimum, maximum, cooldown, and application warm-up behavior.

• Assess whether compatible applications can share a plan without violating isolation, scaling, security, or noisy-neighbor requirements.

• Review registry tier features, geo-replication needs, storage growth, image tagging, and untagged-manifest retention.

• Keep registries network-close to deployment targets where architecture and resilience requirements allow.

Operational safeguards

Do not change production plan capacity or registry retention on cost evidence alone. Validate peak traffic, deployment workflows, image pull behavior, rollback, and owner approval.

• Test autoscale under representative load.

• Before a registry retention policy, verify digest-based pulls and recovery expectations.

• Review forecast and budget impact when changing a tier or scaling envelope.

Evidence and approval model

Decision area	Evidence to collect	Safe action gate
App Service scaling	Request rate, CPU/memory, latency, error rate, worker count, warm-up time	Load-test autoscale; validate min/max and rollback
Plan consolidation	App compatibility, isolation, scale patterns, operational ownership	Architecture and service-owner approval
Registry retention/tier	Image age, tags, pull references, replication and feature requirements	Dependency scan and staged retention policy

Source material

This reference summarizes the following official guidance. Check the linked sources for current service limits, billing terms, and implementation details.

- **Azure App Service cost management**
https://learn.microsoft.com/en-us/azure/app-service/overview-manage-costs

- **Azure Well-Architected App Service guidance**
https://learn.microsoft.com/en-us/azure/well-architected/service-guides/app-service-web-apps

- **Azure Container Registry best practices**
https://learn.microsoft.com/en-us/azure/container-registry/container-registry-best-practices

- **Azure Container Registry retention policy guidance**
https://learn.microsoft.com/en-us/azure/container-registry/container-registry-retention-policy